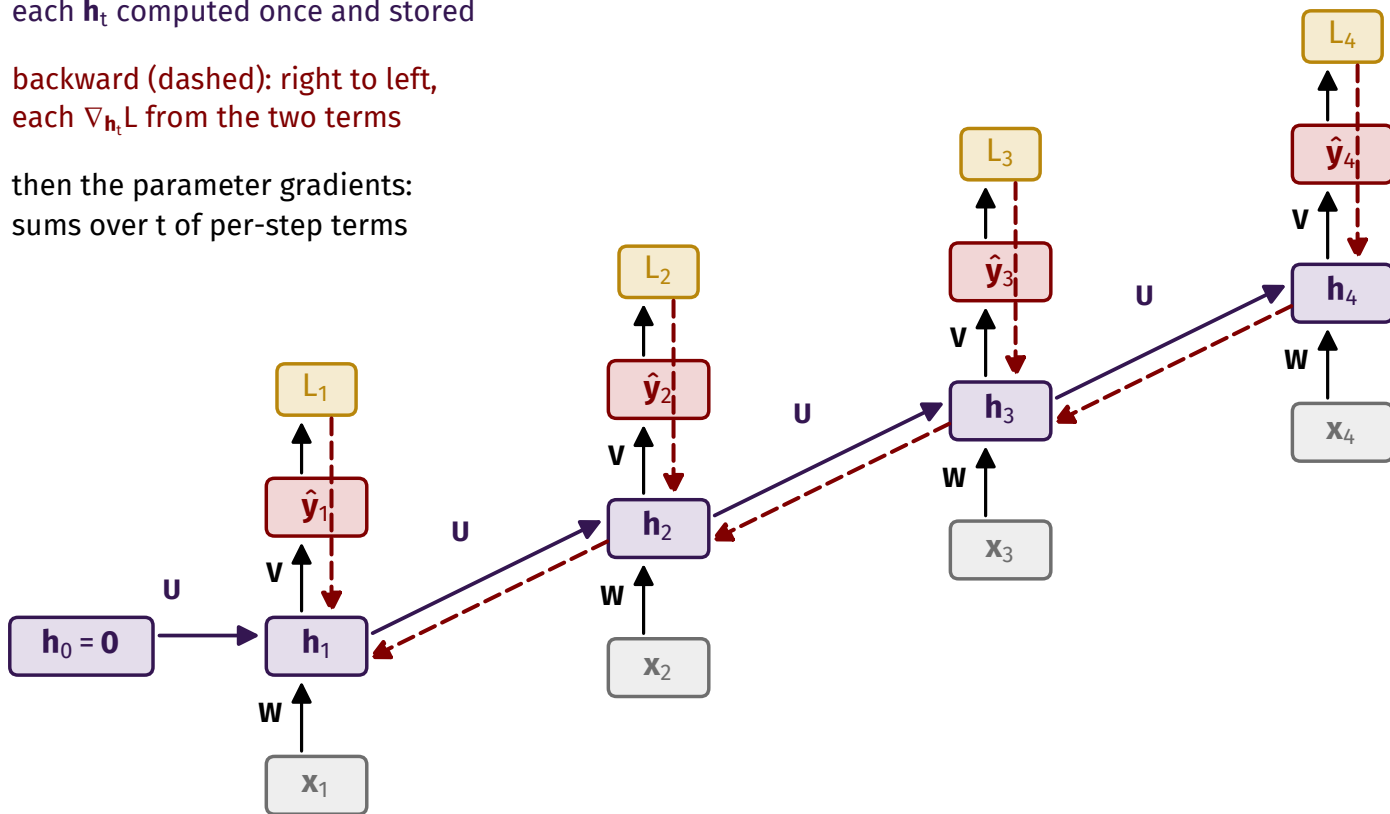


forward (solid): left to right,
each $\mathbf{h}_t$ computed once and stored

backward (dashed): right to left,
each $\nabla_{\mathbf{h}_t} L$ from the two terms

then the parameter gradients:
sums over t of per-step terms



$$\mathbf{V}^\top \nabla_{\mathbf{o}_t} L$$

from the loss at the same step

$$\mathbf{U}^\top \text{diag}(1 - \mathbf{h}_{t+1}^2) \nabla_{\mathbf{h}_{t+1}} L$$

from the next state, back along $\mathbf{U}$
the two add up to $\nabla_{\mathbf{h}_t} L$ (Eq. 10.20)